

Problem Set 2

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Problem 1. Strong Connectivity

(a) In a directed graph G equivalence relation R on the vertices of G as follows: $u R v$ is true IFF there exists a directed walk from u to v and a directed walk from v to u . (Note: A walk from u to v is expressed as $u \rightarrow v$) Now to show that the given relation is an equivalence relation:

- **Reflexive:** By the definition of a walk, there always exists a walk of length 0 from a to a .
- **Symmetric :** If $(u \rightarrow v) \wedge (v \rightarrow u)$ is true, then by the symmetric nature of $\wedge$ swpping the terms does not change the truth value; therefore, $u R v$ implies $v R u$.
- **Transitive:** If $a R b$ and $b R c$ then we know there exists walks $(a \rightarrow b)$, $(b \rightarrow a)$, $(b \rightarrow c)$, $(c \rightarrow b)$ and by concatenating those walk we get $(a \rightarrow b \rightarrow c)$ which implies $(a \rightarrow c)$ and similarly, by doing the same for the other two walks we get $(c \rightarrow a)$, and together they imply $a R c$.

(b) By definition, Relation R directly implies that two vertices are strongly connected, so any two vertices u and v that are in same connected components must satisfy equivalence relation $(u R v)$.

Problem 2. Bijections via Inverses

(a)

Proof by contradiction. Assume f is not injective. Hence, $\exists a, b \in A$ such that $a \neq b$ and $f(a) = f(b)$. Now $g(f(a)) = a$ and $g(f(b)) = b$ by definition but since $f(a) = f(b)$, $g(f(a)) = g(f(b)) = a = b$ which is a contradiction. $\square$

(b)

proof by contradiction. Assume f is not surjective. Hence, $\exists b \in B$ such that $\forall a \in A, f(a) \neq b$. But $f(g(b)) = b$ is true for all $b \in B$ which is a contradiction. $\square$

(c) From parts (a) and (b), we know f is a function, total, injective, and surjective. Hence f is a bijection.

Problem 3. Increasing Correspondence

(a) Let $(a, b, c) \in A$. Now by the definition $f(a, b, c) = (c - b, b - a, a)$ and because $a \leq b \leq c$ we observe that $c - b \geq 0$, $b - a \geq 0$ and $a \geq 0$, which implies that for every valid input pair there is at least one output. Therefore, the given function is a total function.

Now we show that every output of f lies in set B. As we know $\forall (a, b, c) \in A, f(a, b, c) = ((c - b, b - a, a))$, and given pair is part of set B IFF

$$\begin{aligned}(c - b) + 2(b - a) + 3a &= 1000 \\ a + b + c &= 1000\end{aligned}$$

and $(a + b + c = 1000)$ is true since they are in set A, which satisfies all necessary conditions and by this we can conclude that every element $(a, b, c) \in A$, the resulting triple $f((a, b, c))$ lies in B.

(b) Total function $g : B \rightarrow A$, $g(x, y, z) := (z, y + z, x + y + z)$

Now, to show that g is a total function, let $(x, y, z) \in B$. Now by the definition $g(x, y, z) = (z, z + y, x + y + z)$ and because pair (x, y, z) is defined over natural number sums $(y + z)$ and $(x + y + z)$ are deterministic guarantees, which implies that for every valid input pair there is at least one output. Therefore, the given function is a total function.

Now we show that every output of g lies in set A. As we know $\forall (x, y, z) \in B$, $g(x, y, z) = ((z, z + y, x + y + z))$, and $g(x, y, z)$ pair is part of set B IFF

$$z + z + y + x + y + z = 1000$$

$$x + 2y + 3z = 1000$$

and $(x + 2y + 3z = 1000)$ is true since they are in set B, which satisfies all necessary conditions, and by this we can conclude that every element $(x, y, z) \in A$, the resulting triple $f((a, b, c))$ lies in B.

Now to show that $g(f((a, b, c))) = (a, b, c)$ for every $(a, b, c) \in A$ by placing the output of f we get $g((c - b, b - a, a))$ and also applying the value of g we get $g((c - b, b - a, a)) = (a, b, c)$.

Similarly following the same procedure for $f(g(x, y, z))$ we get $f(g(x, y, z)) = f(z, z + y, x + y + z) = x, y, z$ and these conditions are enough to prove that g is an inverse of f .

(c) Since we proved that g is a two-sided inverse of f , and that is necessary and sufficient to prove that f is bijection

Problem 4. Counting Graphs

(a)

$$2^{\frac{n(n-1)}{2}}$$

(b)

$$n \times 4 \times 3 \times 3 = 36n$$

Problem 5. Counting

(a) There are total $21!$ ways to arrange 21 consonants. Now, since every vowel appears to the left of a consonant, we must choose 5 total consonants to make a gap in front and place a vowel. There are $\binom{21}{5}$ ways to make such a gap, and now there are $5!$ ways to put a vowel into those gaps.

Hence total way =

$$21! \times \binom{21}{5} \times 5!$$

(b) We first need to make a total of 5 [VCC] boxes and the remaining 11 consonants in [C] boxes:

$$\frac{\left(\binom{21}{2} \times \binom{19}{2} \times \binom{17}{2} \times \binom{15}{2} \times \binom{13}{2} \times 2^5\right)}{5!} \times 5! \times 16!$$

We needed to divide by $5!$ because the count is for distinct sections of groups, meaning (bc, df,...) is the same as (df, bc,...) and 2^5 because pairs bc and cb are distinct. and there are $5!$ ways to place vowels in each box. and there are a total of 11 boxes of [C], so the number of ways to arrange 16 distinct boxes is $16!$

And simplifying the given equation gives:

$$\frac{21! \times 16!}{11!}$$

(c) (Note: Another concise way to express the equation in (b) is used here, which is via the multinomial coefficient) Need to choose n partitions each of size 2, from $2n$ choices that can be done in:

$$\binom{2n}{2, 2, 2, \dots, 2} \times \frac{1}{n!} = \frac{(2n)!}{2^n n!}$$

We must here, just like in part (b), the binomial coefficient gives an ordered pair, and since there are a total of n terms, there are $n!$ copies of same arrangement.

(d) We can simply apply the Stars and Bars combinatorics theorem, where stars represent the number of times a digit is in a type and there are 8 total digits; also, bars categorize how many times each digit appears, so to create 10 partitions we need 9 bars

$$\binom{\text{total position to place bars}}{\text{bars}} = \binom{\text{stars+bars}}{\text{bars}} = \binom{8+9}{9} = 24,310$$